of trading performance. Such measures are optimized with respect to some mathematical or statistical precedent. However, ultimately our results are analyzed financially to which these measures are not optimized. To whit: the forecast accuracy is calibrated through model estimation but the true value should be based on performance of the trading strategy (Dunis et al., 2004). Common trading performance measures are detailed in Tables 13.4−13.7. Important measures include the Sharpe ratio, maximum drawdown and the average gain/loss ratio. The Sharpe ratio is a risk-adjusted measure of return and allows for direct comparison across the industry. Maximum drawdown is a measure of downside risk, and the average gain/loss ratio is a measure of overall profit (Dunis and Jalilov, 2002; Fernandez-Rodriguez et al., 2000). Back-test measures will shed light on different aspects of the strategy, determining the overall quality of the forecasts, as in the end financial gain depends more on trading performance than forecasting accuracy.

Table 13.4 Trading Simulation Performance Measures

Performance Measure	Description	
Annualized Return	$R^A = 252 \cdot \frac{1}{N}\sum_{t=1}^{N} R_t$	(13.37)
Cumulative Return	$R^C = \sum_{t=1}^{N} R_T$	(13.38)
Annualized Volatility	$\sigma^A = \sqrt{252} \cdot \sqrt{\frac{1}{N-1}\sum_{t=1}^{N}(R_t - \overline{R})^2}$	(13.39)
Sharpe Ratio	$SR = R^A / \sigma^A$	
Maximum Daily Profit	Maximum value of R_t over the period	(13.40)
Maximum Daily Loss	Minimum value of R_t over the period	(13.41)
Maximum DrawDown	Maximum negative value of $\sum(R_t)$ over the period	(13.42)
	$MD = \min_{t=1,\ldots,N}\left(R_t^C - \max_{i=1,\ldots,t}(R_i^C)\right)$	(13.43)
% Winning Trades	$WT = 100\ldots\frac{\sum_{t=1}^{N} F_t}{NT}$ where $F_t = 1$ if Transaction $\text{Profit}_t \geq 0$	(13.44)

Source: Dunis et al. (2004)